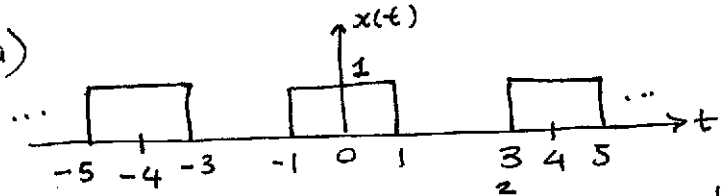
ELEC 361 Midterm Exam Solutions
Fall 2020

① $x(t) = \sum_{n=-\infty}^{+\infty} \Pi\left(\frac{t-4n}{2}\right)$, $h(t) = 8\,\text{Sinc}(2t)\,\text{Cos}(4\pi t)$

a)



$T_0 = 4$, $\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2}$, $f_0 = \frac{1}{T_0} = \frac{1}{4}$

$$x_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt = \frac{1}{4}\int_{-2}^{2} x(t)\,dt = \frac{1}{4}\int_{-1}^{1} 1\,dt = \frac{1}{2}$$

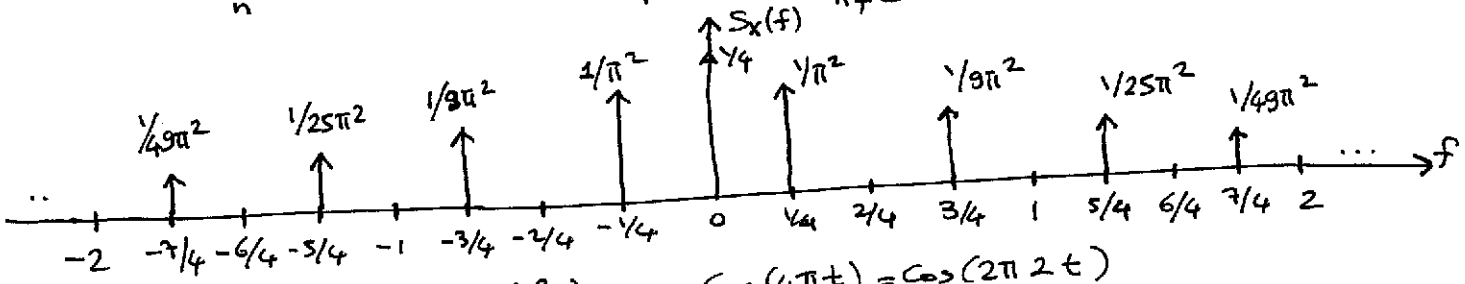
$$x_n = \frac{1}{T_0}\int_{T_0} x(t).\,e^{-jn\omega_0 t}\,dt = \frac{1}{4}\int_{-1}^{1} 1.\,e^{-jn\frac{\pi}{2}t}\,dt = \frac{e^{-jn\frac{\pi}{2}t}}{4\times -jn\frac{\pi}{2}}\Bigg|_{-1}^{+1} =$$

$$= \frac{e^{-jn\frac{\pi}{2}} - e^{+jn\frac{\pi}{2}}}{-2j\cdot n\pi} = \frac{1}{n\pi}\sin\left(n\frac{\pi}{2}\right)$$

$$x(t) = \sum_{n=-\infty}^{+\infty} x_n.\,e^{jn\omega_0 t} = \frac{1}{2} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{+\infty} \frac{1}{n\pi}\sin\left(n\frac{\pi}{2}\right) e^{jn\frac{\pi}{2}t}$$

$$X(f) = \sum_{n=-\infty}^{+\infty} x_n\,\delta(f-nf_0) = \frac{1}{2}\delta(f) + \sum_{\substack{n=-\infty \\ n\neq 0}}^{+\infty} \frac{1}{n\pi}\sin\left(n\frac{\pi}{2}\right)\delta(f-n/4)$$

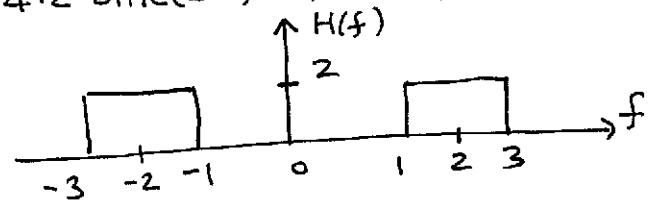
$$S_x(f) = \sum_n |x_n|^2\,\delta(f-nf_0) = \frac{1}{4}\delta(f) + \sum_{n\neq 0}\left[\frac{1}{n\pi}\sin\left(n\frac{\pi}{2}\right)\right]^2\delta(f-n/4)$$



$\text{Cos}(4\pi t) = \text{Cos}(2\pi\, 2t)$

b) $A\tau\,\text{sinc}(t\tau) \xleftrightarrow{\mathcal{F}} A\,\Pi\left(\frac{f}{\tau}\right)$

$4\cdot 2\,\text{Sinc}(2t) \xrightarrow{\mathcal{F}} 4\,\Pi\left(\frac{f}{2}\right) \Rightarrow H(f) = 2\left[\Pi\left(\frac{f-2}{2}\right) + \Pi\left(\frac{f+2}{2}\right)\right]$



ESD

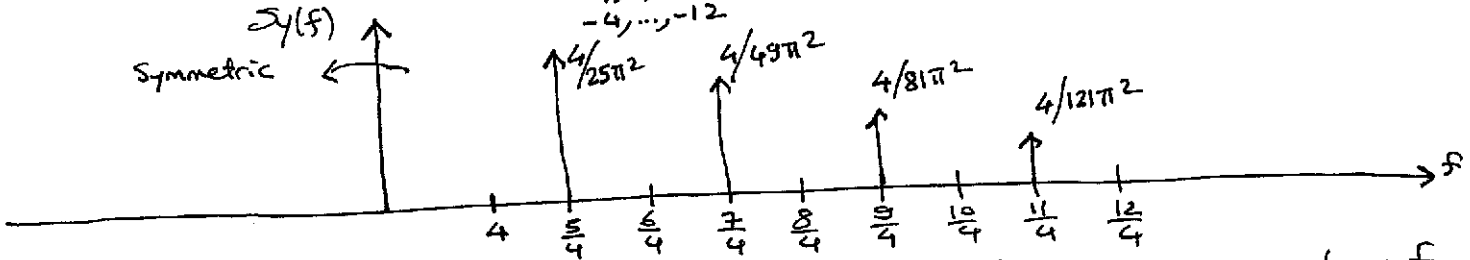
$\mathcal{E}_h(f) = |H(f)|^2$

The same sketch, gain = 4

$$R_h(\tau) = \mathcal{F}^{-1}\{\mathcal{E}_h(f)\} = 16\,\text{Sinc}(2\tau)\,\text{Cos}(4\pi\tau)$$

$$E_h = R_h(0) = 16\,\text{Sinc}(0)\,\text{Cos}(0) = 16\ \text{J}\,, \quad E_h = \int_{-\infty}^{+\infty}\mathcal{E}_h(f)\,df = 2\int_1^3 4\,df = 16\ \text{J}$$

c) $S_y(f) = S_x(f).|H(f)|^2 = 4\sum_{\substack{n=4,\dots,12 \\ -4,\dots,-12}} |x_n|^2\,\delta(f-nf_0)$



d) Not distortionless because it does not pass all frequency components of the input signal. (Distortionless system multiplies the input signal with a constant and introduces some delay.)

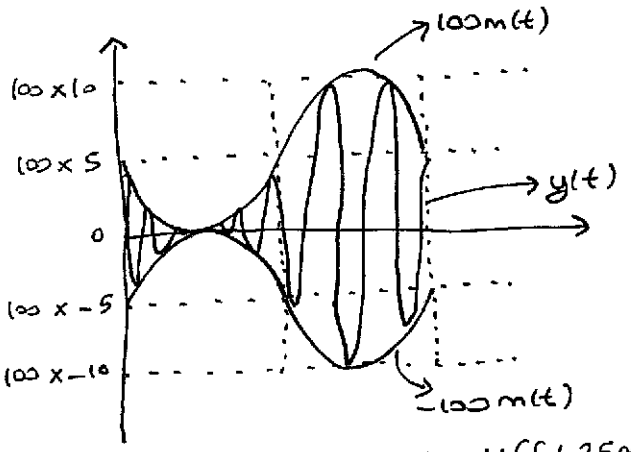
$\forall f\ |H(f)| \neq |K|$

(2) $m(t) = 5 - 5\sin(1000\pi t)$ $\quad c(t) = 100\cos(5000\pi t)$

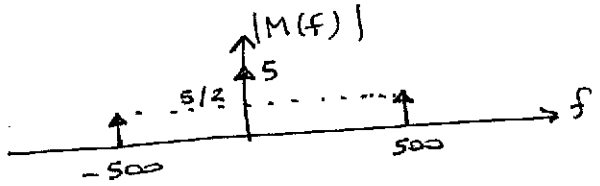
$B = 500$ Hz $\quad A_c \quad f_c = 2500$ Hz

a) $y(t) = m(t).c(t) = 100[5 - 5\sin(1000\pi t)]\cos(5000\pi t)$

$= 500\cos(5000\pi t) - 500.\frac{1}{2}[\sin(4000\pi t) + \sin(6000\pi t)]$

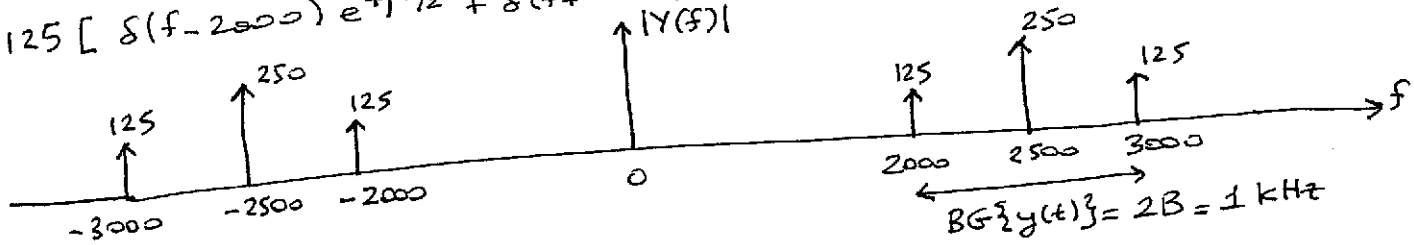


$M(f) = 5\delta(f) + \frac{5}{2}[\delta(f-500)e^{+j\frac{\pi}{2}} + \delta(f+500)e^{-j\frac{\pi}{2}}]$



$Y(f) = 50[M(f-2500) + M(f+2500)] = 250[\delta(f-2500) + \delta(f+2500)] +$

$+125[\delta(f-2000)e^{+j\pi/2} + \delta(f+2000)e^{-j\pi/2} + \delta(f-3000)e^{+j\frac{\pi}{2}} + \delta(f+3000)e^{-j\frac{\pi}{2}}]$



$P_y = \frac{A_c^2}{2} P_m = \frac{100^2}{2}(\underbrace{5^2 + \frac{5^2}{2}}_{37,5}) = 187500$ watt $= 187,5$ kW

b)



$\cos(2\pi(2500+\Delta f)t + \theta)$

$e(t) = y(t).\cos(2\pi(2500+\Delta f)t + \theta) = 100\,m(t)\cos(5000\pi t)\cos(5000\pi t + 2\pi\Delta f t + \theta)$

$= 50\,m(t)[\cos(2\pi\Delta f t + \theta) + \cos(10000\pi t + 2\pi\Delta f t + \theta)]$

$= 50\,m(t)\cos(2\pi\Delta f t + \theta) + 50\,m(t)\cos(10\,\cancel{0}00\pi t + 2\pi\Delta f t + \theta)$ AGT'den geçmez.

$z(t) = 50\,m(t)\cos(2\pi\Delta f t + \theta) \rightarrow P_z = \frac{50^2}{2} P_m = \frac{50^2}{2} \times 37,5 = 46,875$ kW

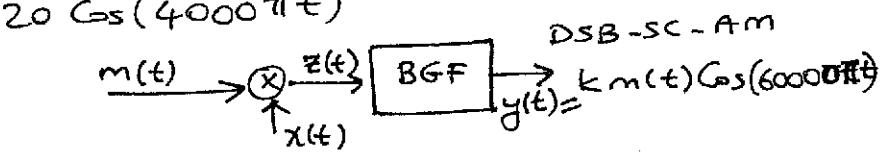
$E(f) = 25[M(f-\Delta f)e^{j\theta} + M(f+\Delta f)e^{-j\theta}] + 25[M(f-f_a)e^{j\theta} + M(f+f_a)e^{-j\theta}]$ $\quad f_a = 5000+\Delta f$

$Z(f) = 25[M(f-\Delta f)e^{j\theta} + M(f+\Delta f)e^{-j\theta}]$

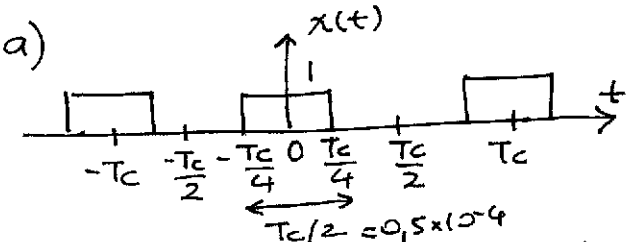
c) No. If $\Delta f = 0$ and $\theta \neq k.\frac{\pi}{2}$; $k \in Z$ and odd, then $m(t)$ can be obtained.

③ $m(t) = 10\cos(2000\pi t) + 20\cos(4000\pi t)$

$x(t) = \sum_n \Pi\left(\frac{t - n\times 10^{-4}}{0{,}5\times 10^{-4}}\right)$



a)



$T_c = 2\times 0{,}5\times 10^{-4} = 10^{-4}$

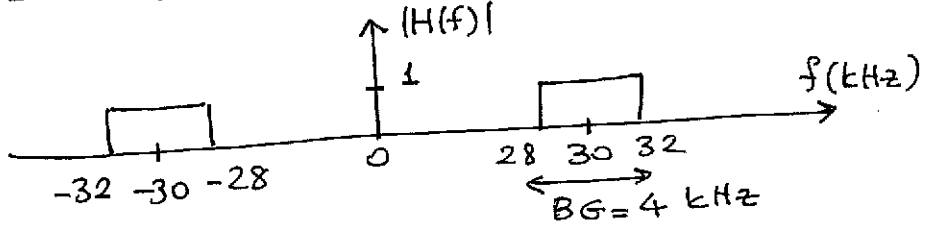
$f_c = \frac{1}{T_c} = 10^4\ Hz = 10\ kHz$

$x(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(2\pi f_c t) - \frac{1}{3}\cos(2\pi 3 f_c t) + \frac{1}{5}\cos(2\pi 5 f_c t) \rightarrow \frac{1}{7}\cos(2\pi 7 f_c t)\dots\right]$

$z(t) = m(t).x(t) \longrightarrow$ BGF, $H(f)$ $\longrightarrow y(t) = k\,m(t).\cos\left(2\pi \frac{30\,000}{3 f_c} t\right) \longrightarrow$ M.F = 30 kHz

$k = \frac{2}{\pi}\times\left(-\frac{1}{3}\right) = \frac{-2}{3\pi}$

$m(t)$: $f_1 = 1\,kHz$, $f_2 = 2\,kHz \Rightarrow BG\{m(t)\} = B = 2\,kHz \Rightarrow BG\{BGF\} = 2B = 4\,kHz$



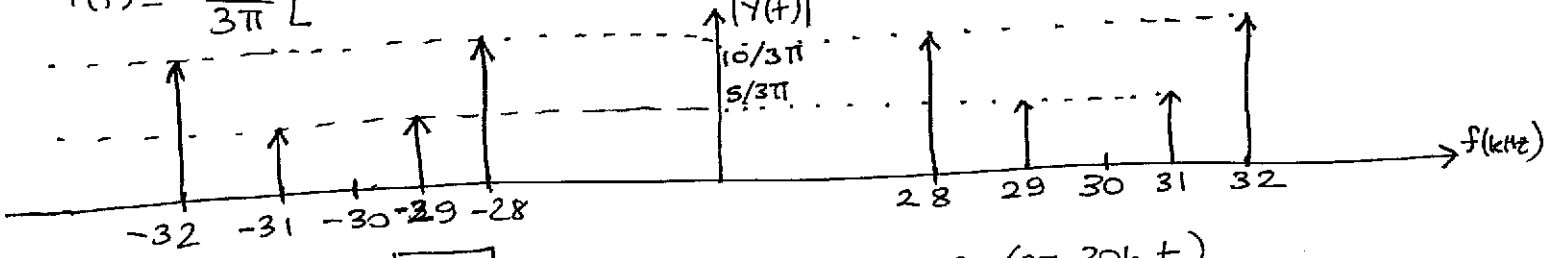
b) $y(t) = \frac{-2}{3\pi} m(t)\cos(2\pi\ 30\,000\ t) \longrightarrow P_y = \frac{(-2/3\pi)^2}{2} P_m$

$P_m = 10^2\frac{1}{2} + 20^2\frac{1}{2} = 250\,W \longrightarrow P_y = \frac{2}{9\pi^2}\times 250 \cong 5{,}63\,W$

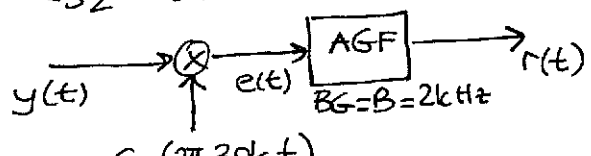
$m(t) = 10\cos(2\pi\ 1k\ t) + 20\cos(2\pi\ 2k\ t)$

$M(f) = 5[\delta(f-1k) + \delta(f+1k)] + 10[\delta(f-2k) + \delta(f+2k)]$

$Y(f) = \frac{-1}{3\pi}\left[M(f-30k) + M(f+30k)\right]$



c)

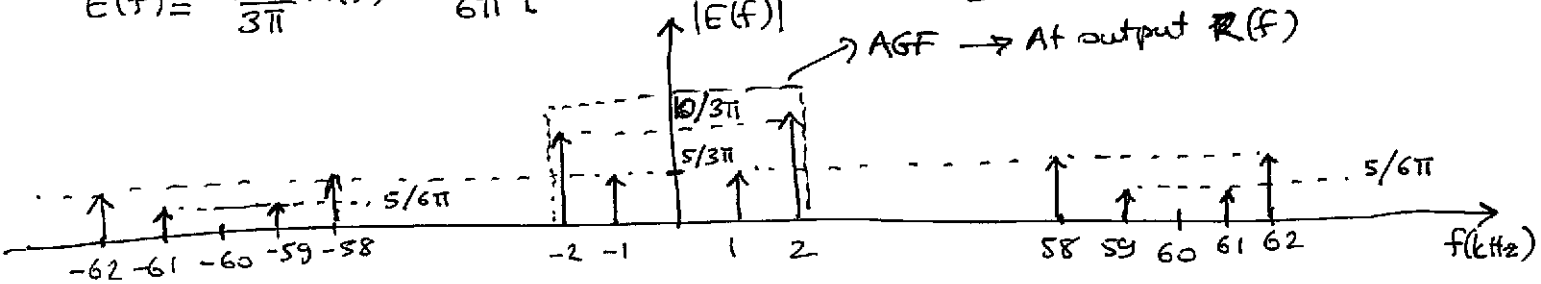


$e(t) = y(t).\cos(2\pi\ 30k\ t)$

$= \frac{-1}{3\pi} m(t)\left[1 + \cos(2\pi\ 60k\ t)\right]$

$r(t) = -\frac{1}{3\pi} m(t) \longrightarrow R(f) = -\frac{1}{3\pi} M(f)$

$E(f) = -\frac{1}{3\pi} M(f) - \frac{1}{6\pi}\left[M(f-60k) + M(f+60k)\right]$



$P_r = \left(\frac{-1}{3\pi}\right)^2 . P_m = \frac{250}{9\pi^2} \cong 2{,}81\,W$